

CHANG DAO, China

WMO: 547510

Lat: 37.9392N

Lon: 120.7256E

Elev: 131

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	22.2	5.2	7.2	24.1	7.5	8.1	25.5	32.3	29.3	30.1	26.1	16.0	340	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	87.5	77.5	85.0	76.0	82.9	75.1	80.1	85.2	78.7	82.9	77.3	81.2	8.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	149.8	83.7	77.3	142.8	81.9	76.0	136.7	80.4	44.1	85.5	42.5	83.2	41.0	81.4	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.2	25.5	22.7	16.8	92.0	3.6	2.4	14.2	93.7	12.0	95.1	10.0	96.4	7.3	98.1		
			14.5	82.0	3.2	2.0	12.2	83.4	10.4	84.5	8.6	85.6	6.2	87.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	30.9	33.4	41.7	52.7	63.0	70.7	76.5	77.2	71.6	61.2	48.0	36.1
	DBStd	17.26	5.23	5.68	6.56	6.69	6.06	4.98	4.50	3.48	3.89	5.52	6.89	5.64
	HDD50	1925	592	464	271	44	0	0	0	0	3	118	433	
	HDD65	4708	1057	884	722	371	113	10	0	4	139	511	897	
	CDD50	3886	0	0	14	126	404	622	823	842	647	351	57	0
	CDD65	1194	0	0	0	3	51	182	358	377	201	22	0	0
	CDH74	6679	0	0	0	13	202	837	2528	2529	557	13	0	0
	CDH80	1316	0	0	0	1	25	151	642	460	37	0	0	0
Wind	WSAvg	11.0	12.6	11.6	11.7	11.8	10.6	9.2	8.8	8.5	9.1	11.2	13.2	13.5
Precipitation	PrecAvg	22.0	0.2	0.4	0.5	1.0	1.9	2.4	6.5	4.9	2.0	1.0	0.9	0.4
	PrecMax	34.7	0.7	1.4	3.0	3.0	4.7	5.9	21.8	9.8	5.3	2.9	3.2	1.3
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.2	0.2	0.8	0.4	0.3	0.0	0.0	0.0
	PrecStd	5.9	0.2	0.4	0.6	0.7	1.1	1.5	4.8	2.5	1.3	0.7	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	52.6	65.3	75.4	83.2	87.3	91.5	90.1	83.6	75.9	66.6	52.5
		MCWB	38.8	43.5	51.9	59.8	65.3	70.8	79.3	80.6	74.7	65.4	56.7	47.6
	2%	DB	42.1	47.5	59.2	70.7	78.8	84.1	88.5	86.6	80.8	72.8	62.2	48.5
		MCWB	37.6	40.8	48.4	57.2	63.2	69.1	78.7	78.9	72.5	64.2	54.4	44.0
	5%	DB	39.7	43.8	54.7	66.8	75.7	81.3	85.8	84.2	79.0	70.4	59.8	45.8
		MCWB	35.6	38.0	45.6	54.5	62.0	68.7	77.7	77.5	71.2	62.6	53.5	41.2
	10%	DB	37.8	41.1	51.0	63.2	72.7	78.9	83.4	82.4	77.2	68.4	57.3	43.5
		MCWB	33.9	36.3	43.1	52.7	60.8	68.1	76.3	76.6	69.8	61.3	51.1	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	44.9	52.9	62.5	68.3	75.6	82.8	82.1	77.4	68.6	58.8	48.3
		MCDB	43.4	50.2	62.6	71.3	78.6	82.2	87.8	87.2	80.3	72.9	63.6	50.9
	2%	WB	38.4	41.4	49.4	58.9	65.8	72.7	80.4	80.4	74.7	66.4	56.4	44.6
		MCDB	41.2	46.3	57.7	67.4	75.8	79.6	86.3	85.2	79.1	70.2	60.7	47.4
	5%	WB	36.4	38.9	46.6	56.3	63.8	71.0	78.9	79.0	72.9	64.6	54.2	41.9
		MCDB	39.4	42.7	53.5	65.0	72.5	77.9	83.8	82.9	77.1	69.1	59.1	45.1
	10%	WB	34.3	37.1	43.8	53.9	62.1	69.6	77.4	77.7	71.4	62.7	51.9	39.7
		MCDB	37.3	40.3	50.0	61.8	70.4	75.9	82.1	81.2	75.6	67.4	56.7	43.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.2	10.0	12.1	11.8	11.3	9.3	7.9	8.1	8.3	7.8	6.8
		MCDBR	9.5	12.2	15.4	17.7	15.9	15.0	12.4	10.0	9.9	10.2	10.5	9.2
		MCWBR	7.4	8.7	9.5	9.9	8.7	6.9	5.9	5.3	6.0	6.8	7.6	8.0
	5% WB	MCDBR	8.9	11.5	14.8	16.8	14.1	12.6	11.0	9.3	8.9	9.5	10.1	9.0
		MCWBR	7.7	8.9	9.7	10.2	8.7	6.8	6.2	5.4	6.2	7.3	8.6	8.6
Clear-Sky Solar Irradiance	taub	0.477	0.532	0.575	0.617	0.647	0.742	0.729	0.658	0.582	0.601	0.550	0.492	
	taud	1.851	1.760	1.703	1.673	1.658	1.543	1.617	1.739	1.830	1.704	1.767	1.841	
	Ebn at Noon	204	211	218	220	216	196	197	208	215	189	179	187	
	Edh at Noon	49	60	70	76	79	88	82	70	61	63	53	46	
All-Sky Solar Radiation	RadAvg	708	972	1393	1648	1828	1722	1521	1485	1360	1046	737	603	
	RadStd	64	89	103	77	144	140	153	148	104	89	92	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+86

Nomenclature: See separate page